

# ABHINAV KAUSHIK

Senior Applied AI Scientist

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## SUMMARY

Senior Applied Scientist with 8+ years applying deep learning, NLP and LLMs to production problems across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI. Fine-tunes transformer models such as cross-encoders, builds RAG and conversational multi-agent applications, and designs automated evaluation to measure quality. Takes work from research and prototyping to deployment on GCP and Azure, collaborating closely with engineers and product stakeholders. Currently building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

## PROFESSIONAL EXPERIENCE

**Senior Applied Scientist** | United Health Group Bengaluru, India | Nov 2025 - Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents and automating the work behind 800 Jira tickets and PRD tasks.
- Engineered a production conversational multi-agent chatbot in LangGraph for AARP medical plans, answering business questions with PySpark analysis, multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Designed automated evaluation for Question Answering and Summary Agents with LLM evaluators, tracing and failure detection, measuring output quality and reducing human-in-the-loop review by 43%.
- Fine-tuned a cross-encoder NLP model on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy.

**Senior Machine Learning Engineer** | Viavi Solutions Chennai, India | Sep 2024 - Nov 2025

- Developed a RAG-based Project Discovery Assistant in LangGraph, combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Applied Sentence-BERT embeddings and DBSCAN clustering in an event-correlation engine tracing network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

**Data Scientist** | Viavi Solutions Bengaluru, India | Nov 2021 - Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

**Data Analyst** | Optum Gurugram, India | Jun 2021 - Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

**Data Engineer** | Nippon Data Systems Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

**Associate Consultant** | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

## PROJECTS

**Think Tree** | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

**Conversational Language Practice Tutor**

- Built a conversational practice tutor for language learners that analyses lesson transcripts with a fine-tuned open model and generates personalised exercises and feedback using RAG.

- Evaluated feedback quality with LLM evaluators and learner-level progress metrics against a baseline, and deployed the PyTorch model behind a FastAPI service on Google Cloud.
- **Technologies:** PyTorch, Hugging Face Transformers, Llama, RAG, NLP, FastAPI, GCP

**RFP Multi-Agent Proposal Automation Platform**

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

**Financial Fraud Detection | Mar 2025**

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

**Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis**

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

**EDUCATION**

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|--------------------------------------------------------------------------------------|---------------------|
| <b>Master of Science, Machine Learning and AI</b>   Liverpool John Moores University | Jan 2021 - Jan 2023 |
| <b>Bachelor of Technology, Information Technology</b>   Jamia Hamdard                | Aug 2013 - Aug 2017 |

**TECHNICAL SKILLS**

**AI / GenAI:** Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract  
**Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning  
**Evals / Observability:** LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis  
**Programming / Data:** Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix  
**MLOps / Cloud:** MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel  
**Data Platforms:** Databricks, Databricks SQL, Apache Spark, Supabase  
**Databases:** TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB  
**Search:** Apache Solr, Lucene Indexing, Information Retrieval  
**BI:** Power BI, QlikSense, SAP HANA

**LANGUAGES**

English: Native

**ADDITIONAL INFORMATION**

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies